Smartphone Usage and Addiction Analysis Explanation

This document explains the updated notebook in plain language: what was done, why each decision was made, how the decisions are backed by data, and how the final decision tree model should be interpreted.

Executive Summary

The analysis predicts `addicted_label` using non-leaking behavioral usage variables from `Smartphone_Usage_And_Addiction_Analysis_7500_Rows.csv`. The corrected analysis keeps all 7,500 rows and fixes an important parsing issue: the value `None` in `addiction_level` is a real category, not missing data.

The final model is a tuned decision tree that excludes identifiers, gender, `addiction_level`, and aggregate/ratio engineered features. It achieved about 93.53% accuracy, 95.48% F1-score, and 91.48% balanced accuracy on the held-out test set. These scores should be interpreted carefully: `addicted_label` is a deterministic binarization of `addiction_level`, so the model is reconstructing a rule-defined target from behavioral features rather than diagnosing real-world smartphone addiction.

1. Analysis Objective

The project goal is to predict whether a row belongs to `addicted_label = 1` or `addicted_label = 0` using behavioral variables such as screen time, social-media hours, gaming hours, sleep hours, notifications, app opens, stress level, and academic/work impact.

The analysis is predictive and educational. It is not clinical. It does not prove that any variable causes addiction, and it should not be used as a medical or diagnostic tool.

Target class	Count	Share
`addicted_label = 0`	2,192	29.23%
`addicted_label = 1`	5,308	70.77%

Main decisions:

- Use `addicted_label` as the binary target.
- Use behavioral usage variables as model inputs.
- Exclude identifiers and answer-key variables.
- Exclude gender from modeling after testing its weak target association.
- Use stratified splitting and class-aware metrics because the target is imbalanced.

2. Corrected Data Loading and Data Quality

The CSV must be loaded with:

```
pd.read_csv(
    "Smartphone_Usage_And_Addiction_Analysis_7500_Rows.csv",
    keep_default_na=False,
    na_values=[""]
)
```

This matters because pandas normally treats the literal text `None` as missing. In this dataset, `None` is a valid `addiction_level` category. If the CSV is loaded with pandas defaults, 819 real `None` rows are incorrectly converted into `NaN`.

Check	Result	Decision
Rows and columns	7,500 rows, 16 columns	Keep full dataset.
Duplicate rows	0	No row removal needed.
Duplicate `transaction_id`	0	Clean identifier, but excluded from modeling.
Duplicate `user_id`	0	Clean identifier, but excluded from modeling.
Actual missing values	0	No missing-value imputation needed.
`addiction_level = "None"`	819 rows	Treat as real lowest-severity category.

3. Addiction Level and Target Definition

The corrected loading shows that `addiction_level` is complete and has four levels: `None`, `Mild`, `Moderate`, and `Severe`.

`addiction_level`	`addicted_label = 0`	`addicted_label = 1`	Interpretation
`None`	819	0	Always label 0
`Mild`	1,373	0	Always label 0
`Moderate`	0	2,874	Always label 1
`Severe`	0	2,434	Always label 1

This means `addicted_label` is a deterministic binarization of `addiction_level`:

- `None` and `Mild` become `addicted_label = 0`.
- `Moderate` and `Severe` become `addicted_label = 1`.

Therefore, `addiction_level` is not an independent predictor. It is effectively the answer key for the target and must be excluded from the model.

4. Gender Decision

Gender is retained for descriptive checks but excluded from modeling. The target rates are close across Female, Male, and Other, and Cramer's V is only about 0.0232, indicating a very weak categorical association.

Gender	Label 0	Label 1	Label 1 share
Female	733	1,728	70.22%
Male	709	1,844	72.23%
Other	750	1,736	69.83%

The project keeps all rows and avoids forcing `Other` into `Male` or `Female`. Excluding gender also keeps the model focused on behavioral usage variables instead of demographics.

5. Preprocessing and Feature Preparation

The model feature matrix excludes:

- `transaction_id` and `user_id`, because identifiers can encourage memorization and do not represent behavior.
- `addicted_label`, because it is the target.
- `addiction_level` and `addiction_level_display`, because they define or directly proxy the target.
- `gender`, because it showed weak association with the target and is not needed for predictive performance.

The notebook creates diagnostic engineered fields:

- `usage_component_total_hours`
- `notifications_per_screen_hour`
- `app_opens_per_screen_hour`
- `sleep_below_7_hours`

The final recommended model excludes the aggregate and ratio engineered fields because later diagnostics show they are not clean behavioral measurements.

6. Exploratory Data Analysis

The strongest descriptive differences are in screen-time-related variables. Users with `addicted_label = 1` have higher average daily screen time, weekend screen time, and social-media hours than users with `addicted_label = 0`.

Variable	Mean for label 0	Mean for label 1	Difference
Daily screen time	5.16 hours	8.47 hours	+3.31 hours
Weekend screen time	6.89 hours	10.21 hours	+3.32 hours
Social-media hours	2.28 hours	3.72 hours	+1.44 hours

Other variables, such as sleep hours, notifications per day, app opens per day, stress level, academic/work impact, and gender, show weaker separation.

7. Effect Sizes

Because the dataset has 7,500 rows, p-values can become significant even for tiny differences. The updated notebook adds rank-biserial effect sizes to show which variables have practical separation between label groups.

Feature	Rank-biserial effect size
`daily_screen_time_hours`	+0.731
`weekend_screen_time`	+0.712
`social_media_hours`	+0.527
`sleep_hours`	+0.045
`gaming_hours`	+0.011

The practical signal is concentrated in daily screen time, weekend screen time, and social-media hours. Most other numeric variables have weak practical separation.

8. Collinearity and Engineered Feature Checks

`daily_screen_time_hours` and `weekend_screen_time` are highly correlated:

Check	Value
Correlation between daily and weekend screen time	0.964

This means they should be interpreted as overlapping screen-time information, not as two independent behavioral drivers.

The notebook also checks whether component-hour fields behave like clean parts of daily screen time. They do not.

Consistency check	Count	Share
`social_media_hours + gaming_hours + work_study_hours` exceeds total daily screen time	4,553	60.71%
`sleep_hours + daily_screen_time_hours + work_study_hours` exceeds 24 hours	156	2.08%

Because the component fields are not mutually exclusive parts of total screen time, `usage_component_total_hours` and the per-screen-hour ratios are treated as diagnostics only. They are excluded from the final recommended model.

9. Modeling Workflow

The notebook uses an 80/20 train-test split with stratification so the train and test sets preserve the target class proportions. Model-family selection is done with 5-fold cross-validation inside the training set, then the selected model is evaluated on the held-out test set.

Models compared:

- Majority-class baseline

- Logistic regression
- Decision tree
- Naive Bayes
- KNN
- Neural network

The decision tree performed best in training-only cross-validation and remains explainable, so it was selected for tuning.

10. Recommended Final Model

The final recommendation is the simpler tuned decision tree without `usage_component_total_hours`, `notifications_per_screen_hour`, or `app_opens_per_screen_hour`.

Metric	Value	Meaning
Accuracy	93.53%	Overall share of correct predictions.
Precision for label 1	94.55%	Of predicted label 1 rows, how many were truly label 1.
Recall for label 1	96.42%	Of actual label 1 rows, how many were found.
Recall for label 0	86.53%	Of actual label 0 rows, how many were found.
Balanced accuracy	91.48%	Average recall across both classes.
F1-score for label 1	95.48%	Balance of precision and recall for label 1.
Macro-F1	92.07%	Average F1 across both classes.

Confusion matrix:

	Predicted 0	Predicted 1
Actual 0	379	59
Actual 1	38	1,024

The model is strong compared with the baseline, but the interpretation must be careful. Since the target is rule-defined from `addiction_level`, the model is learning to reconstruct that binary rule from behavioral variables in this dataset.

11. Decision Log

Decision	Why it was made	Effect on analysis
Load CSV with `keep_default_na=False`	Preserves None as a real category.	Removes fake missing-data narrative.
Keep all gender categories	`Other` is observed, not missing.	Preserves all 7,500 rows.
Exclude gender from modeling	Cramer's V is only about 0.0232.	Keeps model focused on behavior.
Exclude `addiction_level`	It deterministically defines `addicted_label`.	Prevents target leakage.
Add effect sizes	Large samples can make tiny differences significant.	Separates practical signal from statistical noise.
Flag daily/weekend collinearity	Correlation is 0.964.	Avoids treating near-duplicate features as independent drivers.
Exclude aggregate/ratio engineered fields from final model	Component-hour arithmetic is inconsistent.	Improves interpretability and defensibility.
Use training-only CV	Avoids choosing the model based on test data.	Makes held-out reporting cleaner.

12. Limitations

- The dataset appears synthetic or rule-defined, so the findings may not generalize to real populations.
- The target is a deterministic binarization of `addiction_level`, not an independently validated clinical diagnosis.
- The model should not be used as a medical or clinical tool.
- The analysis shows association and prediction, not causation.
- `daily_screen_time_hours` and `weekend_screen_time` are highly collinear.
- Component-hour fields do not behave as clean subparts of total screen time.
- Feature importance values are specific to the selected decision tree and should not be treated as causal importance.

13. Technical Term Glossary

Term	Meaning in this analysis
`addicted_label`	The binary target being predicted.
`addiction_level`	A four-level severity label that defines the binary target.
Baseline model	A simple comparison model, here always predicting the majority class.
Balanced accuracy	Average recall across both classes.
Class imbalance	One target class is much more common than another.
Collinearity	When two input variables carry highly overlapping information.
Confusion matrix	Counts true positives, true negatives, false positives, and false negatives.
Cramer's V	Association strength between categorical variables, from 0 to 1.
Cross-validation	Repeated training-data splits used to estimate model performance.
Decision tree	A model that predicts through if-then split rules.
Effect size	A measure of practical difference or separation between groups.
F1-score	A metric balancing precision and recall.
Feature	An input variable used by the model.
Feature importance	A model-specific score showing which inputs contributed most.
Hold-out test set	Data kept aside for final performance reporting.
Leakage	When answer-like information is accidentally included as input.
Macro-F1	F1 averaged across classes equally.
One-hot encoding	Turning category labels into 0/1 columns.
Precision	Of predicted positives, the share that were truly positive.
Rank-biserial correlation	An effect size for comparing two groups on a numeric or ranked variable.
Recall	Of true positives, the share the model found.
Stratified split	A split that preserves class proportions.
Target proxy	A variable that indirectly or directly gives away the target.